

GUILHERME ELLENA

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Undergraduate researcher and software developer with experience spanning scientific instrumentation, computer vision, machine learning, and fullstack development. Currently pursuing a B.Sc. in Applied and Computational Mathematics at UNICAMP, with active research at CNPEM/LNLS (PIBIC/CNPq), ITA, Recod.ai, and international collaboration through the UNIGOU program. Track record of building production systems for 80+ companies.

EXPERIENCE

Software Developer

[January 2026] — Present

Polo Wear and Ingersoll Rand

Technologies: Machine Learning, Artificial Intelligence, Data Science, Automation, Dashboards

- Development of corporate organization platforms and software, as well as tax systems focused on optimizing operations for large-scale clients.
- Application of advanced Artificial Intelligence, Machine Learning, and Data Science solutions for business process automation and decision-making support.
- Creation of analytical dashboards and structuring of complex data for continuous monitoring and extraction of strategic insights.

FullStack Software Developer CSharp .NET - Automahelp

August 2023 — Present

Technologies: CSharp .NET C# Blazor ASP.NET MSTest MySQL SQL PostgreSQL Syncfusion

- **ERP and Mobile Development:** Built from scratch an ERP for over 80 companies (NFC, financial, and administrative modules) and a fully integrated mobile application (Android and iOS), supported by an API ecosystem and data processing layer.
- **Optimization and Performance:** Implemented architectural improvements that reduced execution time of various processes by 95.75%, lowering operational and memory costs in 24/7 continuous execution systems.
- **Legacy Migration and Modernization:** Led the complete migration of an ERP structured in VB.NET to C# .NET, enhancing software scalability and architecture.
- **Technical Documentation:** Produced detailed knowledge base covering technical documentation, integration examples, and end-user manuals.

FullStack Software Developer at Startup ME2

Feb 2023 — July 2025

Technologies: CSharp .NET Blazor JS HTML CSS Machine Learning with Python NumPy NodeJS MySQL SQL

- Technology architecture and development of a corporate platform serving over six large companies, both international and national. The platform encompasses: large-scale data analysis and collection; ERP development for data administration and processing; integration with services such as Google, YouTube, and Instagram. Additionally, implementation of intelligent solutions focused on predictive modeling to estimate performance indicators, optimize decision-making, maximize value, and reduce costs efficiently.

KNOWLEDGE

Technologies: Experience with CSharp .NET, MSTest, Blazor Server, JavaScript, NodeJS, TypeScript, HTML/CSS, SQL, Express, Handlebars, Bootstrap, NestJS, Prisma, Git, Github, SQL (server and databases), Java, Java Spring Boot, REST Microservices, Cloud Computing: Docker. Databases: MongoDB, MySQL, and NoSQL. ORMs: Sequelize, Mongoose, and Dapper. OS: Linux, Manjaro, Ubuntu, Kali, and POP! OS.

EDUCATION

UNICAMP - Bachelor's Degree in Applied and Computational Mathematics

2024 - 2029

Applied Analysis, Numerical Analysis, Biomathematics, Combinatorics and Number Theory, Mathematical Physics, Computational Geophysics, Computational Methods of Optimization, Operations Research, Mathematical Treatment of Images and Computational Intelligence, Computational Thinking, Algorithms and Computer Programming, Calculus I, Databases, Object-Oriented Programming.

CERTIFICATES

FullStack Developer

2021 - 2025

The Hong Kong University of Science and Technology: Specialization in Software Engineering (J524VAJN41S0) (SDLC) (PDN2MDCDF24N) (7RHM31PKNG68) (1K3G1NH15K7Z) — University of Alberta: Software Architecture — Cloud Computing Docker Training (9F70E70C) — Spread Fullstack Developer (95ABB1D1), Avanade Decola Tech 3rd Edition (772F6D4B), Responsive Web Design - 300 hours FFC — Google Developers Java (1651004288809) — Data Processing Using Python, SQL - Nanjing University (U2TDDTGJS5HB) — IBM Software Engineering (G9DYL5RLNCKC) — University of California, Davis: Data Science (KYQU76WCCHHA) — University of Illinois: Strategic Leadership and Management (PR8CTBNJCTJF)

LANGUAGES

Advanced English; Beginner French; and Portuguese (Native).

HONORS

- **Admissions and Competitive Exams:** UNICAMP: Applied and Computational Mathematics, Statistics. IFSP: Systems Analysis and Development 2024. Called up for EsPCEEx 2024. 1st Place Open Competition at IFSP - Computer Science. 1st Place FMPFM Open Competition - Computer Science. 8th Place UNIVESP - Computer Engineering. 12th Place USP - Exact Sciences. 4th Place UFSCar - Physics. UNESP 9th Place Biosystems Engineering. Admitted twice to EsPCEEx.

EXTRACURRICULAR ACTIVITIES

Phoenix Robotics Team

May 2025 — Present

Department of Computing - UNICAMP

- **Team Coordinator** (Sep 2025 — Present) - Working with machine learning algorithms for virtual line following.
- **Member of the Computing Department** (May 2025 — Present).
- **Achievements:** 1st Place Nationally at "RoboChallenge 2025" - 1st Place Nationally at Copa Pinhão (Category: Line Follower) and qualification for the international finals in Romania (2026).
- *The Phoenix Robotics Team at UNICAMP is an interdisciplinary extracurricular group focused on developing robots for various competitions. Since its founding in 2000, the team has won 27 titles and manages 14 active projects.*

INTERNATIONAL RESEARCH

UNIGOU Training Program 2026 — Undergraduate Researcher

2026

INCBAC Institute — Institute of Czech-Brazilian Academic Cooperation

Czech Republic / Brazil

- Independent research project on the geometric analysis of DNA embeddings produced by pretrained genomic language models (DNABERT, DNABERT-2, HyenaDNA, Caduceus), at the intersection of Applied Mathematics, Deep Learning, and Computational Biology.
- Formalized reverse-complement symmetry as a $Z/2Z$ group action on the sequence space and applied four independent geometric tools — Procrustes disparity, Spearman correlation between pairwise distance matrices, cosine similarity, and linear CKA — to characterize the geometry of the embedding spaces.
- Identified and named the phenomenon of *local consistency without structural consistency*: architectural RC-equivariance is only detectable through structural geometry metrics, not pointwise similarity. Layer-wise analysis further localized where in the network the RC-inconsistency is introduced ($\sim 500\times$ degradation in HyenaDNA).

RESEARCH

Undergraduate Researcher (PIBIC/CNPq)

Feb 2026 — Present

CNPEM — Brazilian Synchrotron Light Laboratory (LNLS), Campinas, SP

- Development of the **CRISP** system (Cryo Ice thickness by Reflection Interferometry and Statistical Processing): non-destructive measurement of vitreous ice thickness in cryo-EM samples using thin-film reflection optical interferometry.
- Combination of multispectral illumination (LEDs at 405, 470, and 528 nm) with physical modeling (Last et al., 2023) for inversion of vitreous ice thickness (20–300 nm, error ± 7 nm), eliminating dependence on the electron beam (EELS/TEM) for this type of measurement.
- Integration of machine learning for real-time thickness map generation, reducing microscope time consumption — a scarce and operationally costly resource.

Undergraduate Researcher

Feb 2026 — Present

Instituto Tecnológico de Aeronáutica (ITA)

- Application of Machine Learning algorithms for modeling water-metal interfaces.

- Research and development at the intersection of Software Engineering and Computer Vision.
- Development of technological solutions to address challenges related to scientific integrity.

Undergraduate Researcher (IC)

August 2025 — May 2026

- Analysis of the **Elastocaloric Effect (eC)** in carbon nanotubes (CNTs) for the development of nanoscale solid-state refrigerators as an ecological alternative to gas compression technologies.
- Application of fully atomistic molecular dynamics (MD) simulations to model the thermal response (ΔT) of CNTs under mechanical deformation (strain).
- Investigation of the efficiency (Coefficient of Performance - COP) of CNTs across a wide temperature range (300–2000 K), simulating Otto-type thermodynamic cycles to validate their viability as versatile refrigerators.

ACADEMIC SERVICE**Reviewer — IEEE International Conference on Image Processing (ICIP 2026)**

Sep 2026

- One of the world's leading conferences in Image Processing and Computer Vision (IEEE Signal Processing Society). Served as a reviewer for workshop-track papers, evaluating technical contributions in image processing, computer vision, and applied machine learning.

OPEN-SOURCE**NASA GeneLab Analysis Working Group (AWG)**

September 2025 — May 2026

- Development of Deep Learning models (CNNs) with TensorFlow and Keras to classify DNA damage in fluorescence microscopy images (Dataset OSD-366).
- Processing of large data volumes (77,000 images) via AWS S3 and metadata analysis with Pandas.
- Focus on bioinformatics and space radiation for long-duration missions (ISS).

CERN ROOT Framework

2025 — Present

- Implementation of the `std::UniformRandomBitGenerator` interface in the `TRandom` class, enabling direct use with STL algorithms such as `std::shuffle` and `std::uniform_int_distribution` (Issue #7979).
- Extension of the TF1 API to support lambdas with `std::size_t` and `std::span` signatures (C++20), via compile-time detection with `if constexpr` (Issue #22071).
- Code reviewed by CERN collaborators; changes include modern C++17/C++20 usage, Doxygen documentation, and test coverage.

CONFERENCES AND EVENTS

- **SEMAP - Scientific Exhibition 2026:** Presentation of the research "Image Duplication Detection in Scientific Publications via Geometric Matching of Local Features – Multimodal Pipeline."
- **SECOMP - Scientific Exhibition 2026:** Presentation of the research "Image Duplication Detection in Scientific Publications via Geometric Matching of Local Features – Multimodal Pipeline."

AWARDS AND RECOGNITION

- **1st Place - UNISOMA Hackathon 2025:** Development of a solution using LLMs and AI to optimize processes for NGOs.
- **1st Place Nationally - RoboChallenge 2025:** Line Follower category (Phoenix Team). Qualified for the world championship in Romania.
- **1st Place Nationally - Copa Pinh o 2025:** Line Follower category. Consolidating national leadership in control algorithms.
- **3rd Place - WeHandle Hackathon 2026:** Event focused on developing LLM/AI technology to solve trust score problems for companies.
- **5th Place Nationally - Gemini / Inteli / USP Hackathon (Medicine):** Development of a technological solution focused on well-being and mental health using Google ecosystem tools. Event promoted by Google in association with UNICAMP (Dec 2025).

- **10th Place Nationally - SKIP + Adapta Challenge Hackathon 2026:** The central challenge was "digitalizing the real economy", identifying operational friction in traditional non-tech sectors and developing functional AI prototypes to address them (Dec 2025).
- **133rd Place Globally - Wunder Fund Market State Forecasting Challenge:** Acted as *Quantitative Researcher* developing a Machine Learning model to predict market states in High-Frequency Trading (HFT) scenarios, with optimized inference for efficient CPU execution and low latency (Dec 2025).

JUNIOR ENTERPRISE

Quanta Jr. - UNICAMP Exact Sciences Junior Enterprise

Feb 2025 — Dec 2025

Machine Learning & AI Agents Analyst

- **Keywords:** Data Science, Machine Learning, Artificial Intelligence, AI Agents, Predictive Modeling, Python, Project Management.
- Worked on Data Science consulting projects, applying predictive models and AI agents to solve real client problems.
- Responsible for project management and technical interface with stakeholders.

SPECIAL TRAINING

- **NB4 — Biological Safety Level 4 (High Containment):** Training in high-containment biosafety procedures, issued by CNPEM. 2026
- **LN2 — Liquid Nitrogen Handling:** Certification in safe handling and storage of liquid nitrogen, issued by CNPEM. 2026

PERSONAL PROJECTS

AR Fluid Simulation - Computer Vision and CFD

Dec 2025 — Present

[GitHub Repository](#) — [Documentation](#)

- Project combining Computer Vision and Computational Fluid Dynamics (CFD) to create an interactive particle physics experience controlled by hand gestures.
- Uses a webcam to track the user's hands in 3D, enabling direct manipulation of a "virtual container" where a liquid simulates real physical properties such as viscosity, pressure, and gravity.

ImageVisLab - Real-Time Image Processing

Dec 2025 — Present

[GitHub Repository](#)

- Development of a tool featuring 23 algorithms across 6 categories (Point Operations, Convolution, Morphology, Edge Detection, Color, and Fast Fourier Transform - FFT).
- Implementation of real-time histogram comparison, Pixel Inspector with distance metrics (Manhattan, Chebyshev, and Euclidean), Animation Mode for step-by-step visualization, and statistical extraction (Mean, Standard Deviation, and Entropy).